

Technical Checkpoint Report

Project: Neural Network based Autonomous Simulation (NN-Auto-Car)

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1. Project Overview

This capstone project focuses on the development of a high-fidelity **Autonomous Vehicle Simulator**. The objective is to implement a self-driving system capable of navigating complex virtual environments using environmental sensors (Lidar) and neuroevolutionary learning algorithms.

2. Technical Architecture

2.1 Neural Network Structure (MLP)

The vehicle's decision-making engine is a **Multi-Layer Perceptron (MLP)** implemented in `brainModel.js`.

- Input Layer:** 5 neurons corresponding to the Lidar sensor array's return signals.
- Hidden Layer:** 6 neurons with step-function activation.
- Output Layer:** 4 neurons mapped to `Forward`, `Left`, `Right`, and `Reverse`.

2.2 Environmental Perception (Lidar Array)

The perception system (`lidarArray.js`) simulates a 5-beam sensor spread with a $\pi/2$ field of view. Data is normalized into distance offsets (0 to 1), serving as direct inputs to the neural state.

2.3 Physics & Collision Engine

A custom physics engine handles momentum, friction, and steering angles. Collision detection utilizes polygon-to-segment intersection algorithms for damage assessment.

3. Training Methodology: Neuroevolution

The project utilizes **Genetic Algorithms** for population-based learning:

- Swarm Intelligence:** Population size of 100 vehicles per generation.
- Mutation Logic:** The "Champion Brain" is preserved and mutated at a 10% rate for subsequent generations.
- Survival Scoring:** Entities are ranked based on velocity and duration without collision.

4. Current Progress & Milestones

- ✓ Core Physics Engine and coordinate translation.
- ✓ Sensor Fusion and visual debugging layers.
- ✓ Brain Persistence and localStorage optimization.
- ✓ Live Neuro-Visualization of hidden layer activations.

5. Next Steps

- Dynamic Traffic Generation:** Adding stochastic paths for dummy vehicles.
- Global Pathfinding:** Integrating `nodeGraph.js` for nav-mesh coordination.
- Architecture Scaling:** Expanding the hidden layer depth for improved generalization.